

Problem Set 3



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Advanced Methods in Applied Statistics
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Info

- The submission is:
 - A write-up as a PDF document, which includes any plots, diagrams, tables, pictures, and explanations
 - In a separate “file”, submit all code used to derive the results
 - Tarball, zipped directory, lots of individual files w/ self-explanatory titles, etc.
 - Do NOT include lines of code in your write-up. If results are dependent on coding choices then include those comments in the write-up.
- Include any original data files or how the data was accessed
 - If you use a internet scraping tool, note the date when you retrieved the data
 - If you can save the data to a file, do so and submit the data file. There is no need to change the format, e.g. HTML, XML, txt, JSON...

Problem 1 (3 pts. Total)

- Census data collected in the 1990s of working adults in many countries can be used as a data set to establish earning potential
- Create a classifier which separates lower income earners ($\leq 50k$) from higher income earners ($> 50k$)
 - See further criteria for training requirements on the next slides (Problem 1a has 2 slides which should be read through entirely before starting)
- The data set has been divided:
 - Training/Testing data set is at https://www.nbi.dk/~koskinen/Teaching/data/earning_potential_train_test_AMAS_2026.txt
 - The analysis data set is at https://www.nbi.dk/~koskinen/Teaching/data/earning_potential_real_2026.txt
 - This particular data file is only used in problem *part c*
 - Include **both input files** when submitting your solution

Problem 1a (1 pt.)

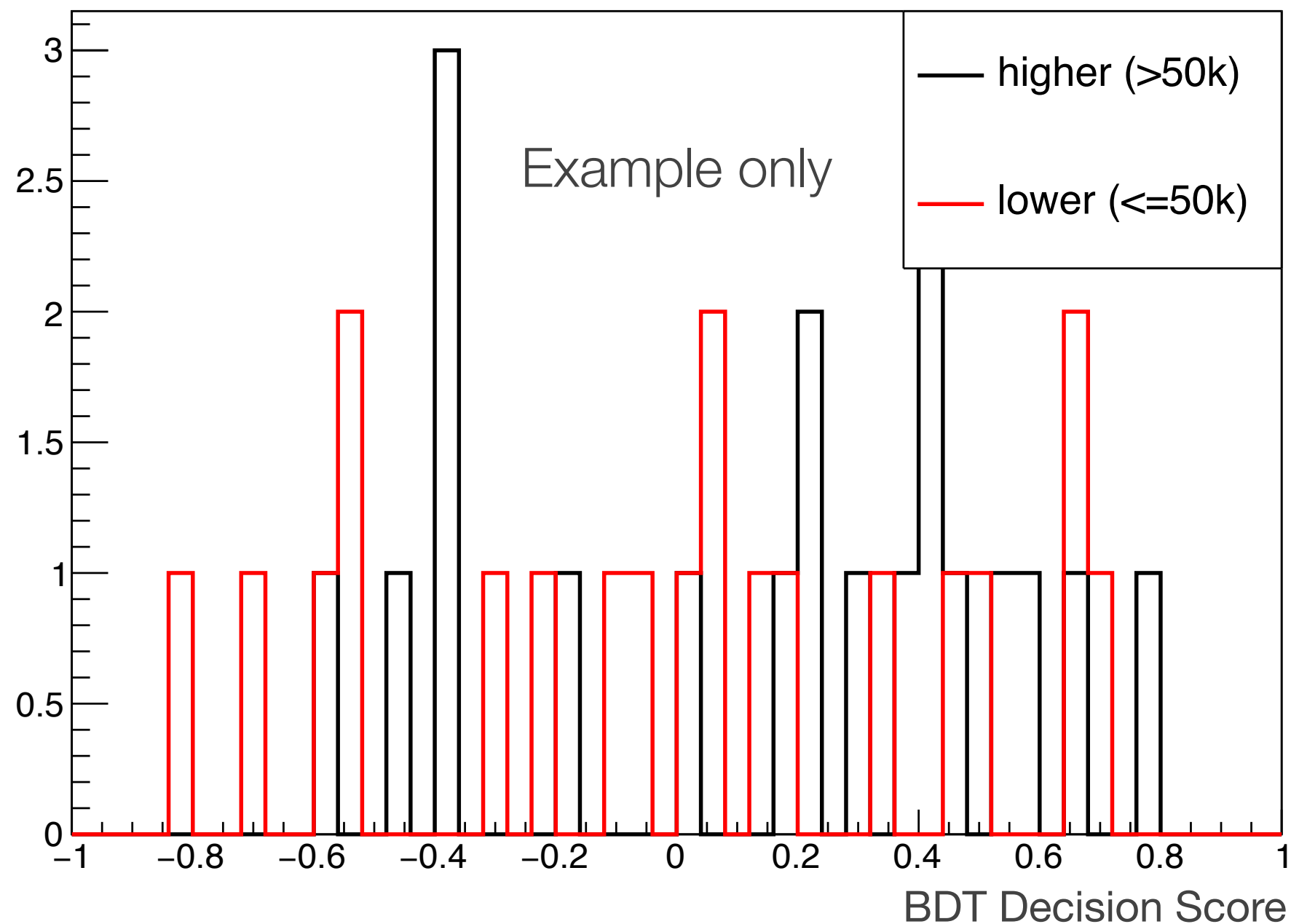
- Using your classifier, what is the **classification accuracy** of the your classification algorithm?
 - Ensure there are at least 500 low earners **and** 500 high earners in a validation/test data set that are **NOT** included in the training data set
 - https://en.wikipedia.org/wiki/Confusion_matrix
 - Classification accuracy is $\frac{TP + TN}{P + N}$
- N.b., that you'll have to use **this** trained algorithm on an unknown data set in *part c*. So be mindful about overtraining for *part a* only to get penalized in *part c* when you have to use the algorithm on unknown data.

Problem 1a (cont.)

- Make a histogram plot using **at least** 500 low earners and **at least** 500 high earners from the train_test.txt file as a function of their decision score.
 - Use 500 low earners and 500 high earners that were **not** part of the training sample
 - Separate the two populations and plot the high earners in **black** and low earners in **red**
 - If your method does not have a 'decision score' then plot each earner as a function of the test statistic which is used for the classification

Problem 1a (example)

- Example here is shown for only 20 entries



Problem 1b (0.5 pts.)

- Report the fraction of high earners in the training data sample, and report the fraction of low earners in the training data set.
 - Keep in mind that the training sample excludes those earners who are part of the validation sample
- Discuss how to identify and avoid overtraining supervised machine learning algorithms and what checks you made to ensure that your trained algorithm does not exhibit signs of overtraining.

Problem 1c (1.5 pts.)

- Using the same classifier developed in Problem 1a, run the classifier over all the entries on the blind/real sample (earning_potential_real_2026.txt)
 - The new data file has an additional first column which is the ID number of the earner
 - Produce a text file which contains **only** the IDs which your classifier classifies as **low earners** (your_name.low_ID.txt)
 - Produce a text file which contains **only** the IDs which your classifier classifies as **high earners** (your_name.high_ID.txt)
 - Basic text files. No Microsoft Word documents, Adobe PDF, or any other extraneous text editor formats. Only a single ID number per line in the text file that can be easily read by `numpy.loadtxt()`.
 - One entry per line and no commas, brackets, parenthesis, etc.

Problem 2 (4 pts. Total)

- There is data regarding crashes in the town of Cary, North Carolina, U.S.A. over the span of years
 - Presumably they are car crashes
 - We will be using specific variables in the data
 - Longitude="lon", Latitude="lat", and the time/data field is "crash_date"
 - Negative longitude implies 'West', whereas positive implies 'East' from a spot in Greenwich Park in London, England
- <https://www.nbi.dk/~koskinen/Teaching/data/cpd-crash-incidents.csv>
 - Use all the data in the file

Problem 2a (1.5 pts.)

- Create a scatter plot of ALL the crash entries as a function of the latitude and longitude (in decimal degrees)
 - https://en.wikipedia.org/wiki/Decimal_degrees
 - Longitude on x-axis and latitude on y-axis
- Make a histogram of the time of day of each crash where the x-axis goes from 0-24 hours.
 - Bin width is 1 hour
 - Lowest bin edge starts at 00:00 (for the HH:MM time format)
- Describe how would you create a kernel density estimation using a gaussian kernel with a bandwidth of 0.25 hours to produce a probability density function of the time of day for crashes
 - This is a qualitative description. Feel free to use written text, diagrams, hand-drawn plots, or any other illustrative tool to describe the KDE PDF generation.
 - N.B. For a 24-hour clock, time 'wraps' at 00:00 & 24:00. For example, 00:01 and 23:59 are very close to each other in actual time, but not numerically.

Problem 2b (1 pt.)

- Create a kernel density estimation using an Epanechnikov kernel with a bandwidth of 0.8 hours to produce a probability density function of the time of day for crashes
- Make a plot of the PDF constructed by the KDE
- In a table in the write-up, provide the evaluation of the PDF at the following times [00:23, 01:49, 08:12, 15:55, 18:02, 21:12, 23:44]
- Using only the time KDE PDF, if additional police officers patrolling the roads reduced the relative crash rates by 10% for a duration of 2 continuous hours, what 2-hour window would be the best to patrol? How much would the 24-hour percentage of crashes be reduced for your choice of 2-hour window of additional patrols?
 - Written slightly differently: the KDE PDF describes the relative crash likelihood during a 24-hour window. If additional police patrolled the entire 24-hour time, then there would be a 10% decrease in the likelihood of accidents. So what is the percentage of the 24-hour daily crashes that will be reduced by your 2-hour window?

Problem 2c (1.5 pt.)

- Create a 2-dimensional KDE PDF from the crash data using the latitude and longitude with an Epanechnikov kernel and a bandwidth of 0.01 in both dimensions
 - This is time independent, so include all times
 - Make a contour plot of the KDE PDF with the longitude on the x-axis and latitude on the y-axis. Include an appropriate color bar for the plot which shows numerical value of the color or include labels on plotted contour lines.
- What is the total percentage of crashes estimated by the KDE PDF to be within the 'box' of longitude range of $[-78.76, -78.72]$ and latitude range of $[35.74, 35.78]$

Problem 3 (1 pt. Total)

- A pace-maker is generated by assembly facilities in 5 different countries with the following defective rates per factory and total worldwide production percentage

Facility	Total % produced	% Defective
A_1	35	2
A_2	15	4
A_3	5	10
A_4	20	3.5
A_5	25	3.1

Problem 3a (0.5 pts.)

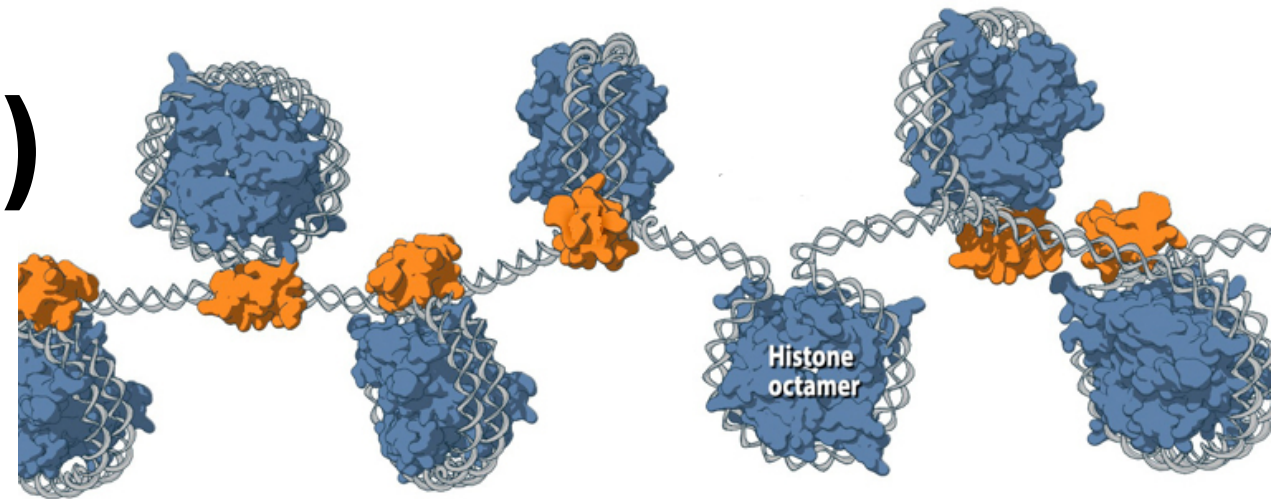
- The world-wide distribution of the actual pace-makers is decoupled from the proximity to the production facilities
- Suppose a pace-maker is found to be defective (D), what is the probability it came from the A_2 facility?
- If a defective pace-maker is found, which facility is it most likely to be from?

Problem 3b (0.5 pts.)

- The CEO of Slightly Evil Inc. wants to ensure that all the facilities have the same probability of being identified with a failed pace-maker, but still wants the least total defects and no changes to the per facility production. How must the percentage of defects change from each facility to accommodate this goal?
 - Submit a list of the altered/updated defective rates for each facility
 - N.b., defective product rates can only increase

```
W/ defective, the updated defective rate for A1: XXXX
W/ defective, the updated defective rate for A2: XXXX
W/ defective, the updated defective rate for A3: XXXX
W/ defective, the updated defective rate for A4: XXXX
W/ defective, the updated defective rate for A5: XXXX
```

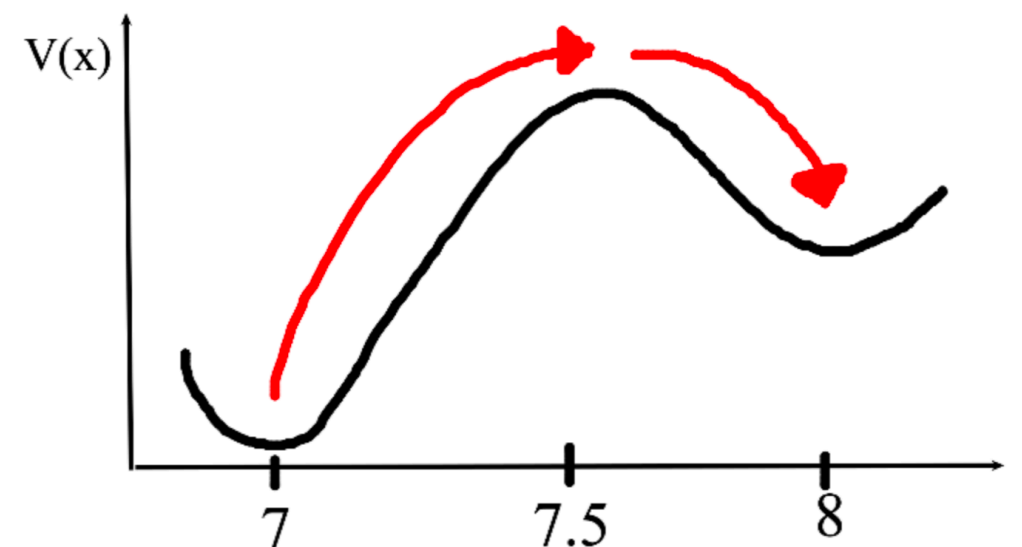
Problem 4* (2 pts total)



- DNA in the nucleus of a cell is wrapped around histones (proteins that DNA can attach to) which we call nucleosomes.
- This wrapping creates a stable configuration, yet experiments have shown that nucleosomes periodically switch between positions by shortly detaching the DNA and taking a step.
- In the following dataset, we have a landscape of the potential for the nucleosome as well as their positions (integer values) and the energy configuration when detached (half-integer values)
- https://www.nbi.dk/~koskinen/Teaching/AdvancedMethodsInAppliedStatistics2026/data/Remodeler_Landscape.csv

Problem 4a (1 pt.)

- Write, or import, an appropriate sampling mechanism (e.g. Transformation Method for MC, Markov Chains) to solve for the probability distribution of the nucleosome position in the potential landscape for the following scenario:
 - Assume boundaries are periodic
 - Use $P(x) = e^{-V(x) \cdot \beta}$ with $V(x)$ the energy at position x and β a temperature dependent scaling factor
 - When a step to the left or right happens to a half-integer value, it will skip that position and land in the valley behind it.



Problem 4a (cont.)

- Use three different values for $\beta \in [0.01, 0.1, 1]$, create a plot that demonstrates convergence of your obtained distribution and discuss under which conditions the results from such a Markov chain would be valid.
- Compare your results with the analytic estimate obtained from estimating the relative probability of **the valleys** $P(x + 1)/P(x) = e^{-\Delta V \cdot \beta}$ and normalising.
 - The ‘Delta’ symbol (Δ) denotes the difference in values of the potential $V()$.
- Create a plot that allows you to visualise the time step distribution for different positions of your nucleosome. You are free to decide how to visualise this as long as we can see the median and a confidence interval at these positions.

Problem 4b (1 pt.)

- Continue by using $\beta=0.25$
- Extend your sampling algorithm by introducing a remodeller that can randomly attach with the following properties:
 - Attaches and detaches with a rate τ
 - Selects a movement direction (left or right) with equal probability and causes movement to only be possible in that direction until it detaches
- Choose an appropriate value for the rate τ such that we expect 25 steps to have occurred before attaching. Please make clear which rate you have used.

Problem 4b (cont.)

- Graph the updated distribution obtained by introducing a remodeller, make sure to still verify or discuss convergence for all your distributions.
- What does the distribution look like if we change the rate τ ?
- How about if the remodeller was heavily biased in a direction, such as always to the left or to the right. Does the distribution change significantly? Discuss how you came to your conclusions